

Shadowed Speech: an Audio Feedback System which Slows Down Speech Rate

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ABSTRACT

In oral communication, it is important to speak at a speed appropriate for the situation. However, we need a lot of training in order to control our speech rate as intended. This paper proposes a speech rate control system which enables the user to speak at a pace closer to the target rate using Delayed Auditory Feedback (DAF). We implement a prototype and confirm that the proposed system can slow down the speech rate when the user speaks too fast without giving any instructions to the speaker on how to respond to the audio feedback.

CCS CONCEPTS

• Human-centered computing → Sound-based input / output.

KEYWORDS

Computer-Enhanced Interaction, Delayed Auditory Feedback

ACM Reference Format:

Mariko Chiba, Wataru Yamada, and Keiichi Ochiai. 2022. Shadowed Speech: an Audio Feedback System which Slows Down Speech Rate. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3526114.3558640>

1 INTRODUCTION

Speech rate is one of the most important components of a smooth oral communication. It is known that an appropriate speed of speaking changes depending on the situation. For example, the average speech rate of Japanese broadcasted programs ranges from 450 to 570 morae per minute [6], while the speech rates of 320 and 360 morae per minute are reported to be most preferred by non-native speakers of Japanese [8]. Since speaking at an inappropriate speed will lead to communication discrepancies, we need to adjust the speech rate considering the situation.

Speaking at an adequate speed is, however, a difficult task for speakers because situational factors such as the mental state may alter their own speech rate unconsciously. Thus, we need a lot of training in order to control our speech rate as intended. This paper aims to allow the users to control their speech rate with minimal conscious adjustment.

Many studies on speech supporting systems have been reported. Damian et al. proposed a system that provides realtime visual feedback on one's behavior during public speaking [3], and Bubel et al. proposed a device that provides visual and haptic feedback on the voice pitch, filler words, and speech rate during presentation practices [1]. Although these systems increase the users' awareness of their own style of speaking, the users need to control their way of speaking consciously to improve the presentation. Costa et al. proposed an approach for emotion regulation using manipulated audio feedback which doesn't require conscious attention [2]. They reported that participants who received voice feedback with manipulation felt emphasized self-perception without noticing the manipulation of feedback.

We propose a system which aims to control the user's speech rate without their intentional adjustment by means of audio feedback with adaptively changing delay times. This is inspired by SpeechJammer[4] using Delayed Auditory Feedback (DAF): a technique of auditory feedback known to have effect on speech production, in which speakers listen through headphones to their own speech with a short time delay. SpeechJammer uses this phenomenon in order to disturb the speech of others. By contrast, our approach use it to support the improvement of speech in terms of speed using audio feedback with a delay time decided depending on the speaker's speech rate.

We consider that our proposed system will contribute to speech rate optimization in oral communication. For instance, a presentation speaker can speak at an adequate speed by setting the target speech rate in advance. Of more interest is that an audience can adjust the speaker's speech rate by sending speed requests like changing the playback speed of video contents.

2 IMPLEMENTATION

Figure 1a shows the prototype system's architecture. The prototype system consists of a single-sided headset (SENNHEISER SC 45 USB MS) and an audio processing device (MacBook Pro M1 2020). The speaker's voice is input from the microphone on the headset, and the audio feedback is played from the headphone. We adopt a single-sided headset so that the user can hear both direct voice and returned voice from the system.

The input voice is processed in three steps. First, the speech rate is calculated in realtime using the speech recognition results and the timestamps of voice activity detection. We use Julius [5] as speech recognition decoder software. Then, the delay time of audio feedback is determined by the function shown in Figure 1b. We adopt a simple monotonically increasing function in which a delay time is added only if the average speech rate of the last three utterances exceeds the threshold. The slope of the line is set as

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UIST '22 Adjunct, October 29–November 2, 2022, Bend, OR, USA

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ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3558640>

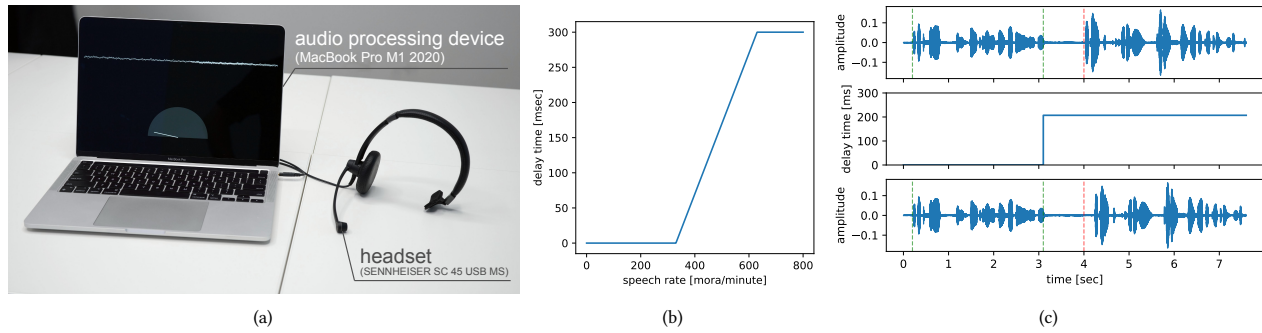


Figure 1: The architecture and the output of the system: (a) The prototype system architecture. The microphone receives the speaker’s voice and the voice is returned with a delay from the headphone. (b) The relation between the speech rate of the speaker and the delay time added to the feedback. If the speech rate exceeds the target speech rate, the delay corresponding to the excess speed is added to the feedback. (c) The input speech waveform and the feedback waveform of the prototype system. The amount of delay time changes while the speaker is speaking.

variable α ; the smaller α is, the shorter the delay time of feedback becomes. Finally, the system produces the audio feedback to which the delay is applied.

The input and output of the system is shown in Figure 1c. The user hears their voice with no delay when speaking slower than the target speed, while they hear delayed feedback when speaking faster than the target speed.

3 EVALUATION

We conducted a user study to confirm whether the proposed system can affect the speech rate of the speaker. We recruited six people (three males and three females) with the average age of 25.83 who agreed to participate in the experiment. They were asked to read presented text in three audio feedback conditions the system provides: no delay, short-time delay ($\alpha = 1$) and long-time delay ($\alpha = 2$). The target speech rate was set to 330 morae per minute at which native Japanese speakers would feel a little slow, and the upper limit of delay time was set to 300 msec. Three patterns of text were excerpted from a Japanese famous novel *Kokoro* [7], with the count of phones in each text pattern ranging from 507 to 576.

Prior to each session, the participants were given instructions of the study. They were asked to read a pattern of text silently, and then to read the same text aloud as if they had been reading it in front of multiple people. This trial was repeated until they finished reading six times (twice for each pattern) under the conditions applied in a random order. Note that the core purpose and the operational details of the audio feedback system were hidden to the participants.

As Figure 2a shows, the average speech rate with the proposed audio feedback becomes slower compared with no delayed audio feedback ($p < 0.05$) although no significant difference is observed between the two conditions of delayed feedback ($\alpha = 1$, $\alpha = 2$). Looking at each user shown in Figure 2b, the difference tends to be larger for the participants who speak fast under no delayed feedback condition (P1 and P5). This suggests that adapting a delay affects fast speakers more strongly.

There are two major limitations of this prototype system. One is that it does not consider individual differences in the susceptibility

to DAF; some users can be affected too strongly to speak smoothly. It is necessary therefore to design a method to decide delay times more intelligently, factoring in such effects on each user. The other is that our approach cannot increase the speech rate because we use a phenomenon that slows down the speed of speaking. We should also consider a way to increase the speed in order to optimize the speech rate in the truest sense.

4 CONCLUSION

We have proposed an audio feedback system which slows down the speech rate using DAF. Our approach is to change delay times depending on the speed of speech in order to improve the speech rate without the speaker’s intentional control. We conducted a user study which confirmed the effects of audio feedback in changing delay times corresponding to the speech rate. The result suggests that the proposed feedback system can slow down the speech of persons speaking at a high rate.

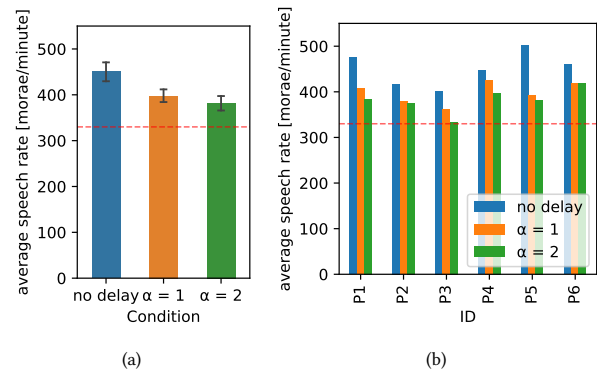


Figure 2: The result of the user study: (a) The average speech rate in each condition. The error bars indicate 95% Confidence Interval (CI). (b) The average speech rate of each speaker. Values closer to the target (330 morae per minute) are better.

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